Paper -1 (Adv-2022-P1-Model)

FST-06

KEY SHEET

MATHEMATICS

1	**40.49**	2	**1.50**	3	**71**	4	**6**	5	**0**	6	**1**
7	**6**	8	**8**	9	**AB**	10	**ACD**	11	**B**	12	**BC**
13	**ACD**	14	**AB**	15	**A**	16	**B**	17	**B**	18	**D**

PHYSICS

19	**3.33**	20	**2**	21	**10.50**	22	**8**	23	**420**	24	**54**
25	**6.60**	26	**1.40**	27	**AC**	28	**AC**	29	**AB**	30	**ABCD**
31	**C**	32	**B**	33	**A**	34	**C**	35	**B**	36	**D**

CHEMISTRY

37	**30**	38	**169**	39	**80**	40	**7**	41	**1**	42	**56**
43	**1.70 to 1.73**	44	**5**	45	**ACD**	46	**BCD**	47	**ABC**	48	**CD**
49	**ACD**	50	**ABC**	51	**D**	52	**A**	53	**B**	54	**C**

SOLUTIONS
MATHEMATICS

01. $I_{n+1}-I_n=\int_0^{\pi}\frac{\sin(2n+1)\theta}{\sin\theta}\,d\theta=\pi\ \forall n\in N$, $I_1,I_2,I_3,.....$ form an AP with common difference π $\quad I_n=\pi+(n-1)\pi\Rightarrow I_n=n\pi$ $\quad \frac{(I_{2024}+I_{2025})}{100\pi}=\frac{4049}{100}=40.49$

02. Mean height of 36 students is $150.5cm\Rightarrow$ sum of all heights is $5418cm$ Given, height of one student was wrongly copied as $165cm$ instead of $156cm$

$\Rightarrow$ Actual sum of heights is $5409cm$, So, Actual mean height is $\frac{5409}{36}=150.25cm$

03. $P(r)=\lambda r\Rightarrow\sum_{r=1}^{6}P(r)=1\Rightarrow 21\lambda=1\Rightarrow\lambda=\frac{1}{21}$

Required Probability $=\sum_{r=1}^{6}\lambda r\frac{(7-r)}{21}=\frac{1}{21^2}\left(\frac{7\times6\times7}{2}-\frac{6\times7\times13}{6}\right)=\frac{8}{63}$

04. $\sum_{k=1}^{n}\frac{k^2-\frac{1}{2}}{k^4+\frac{1}{4}}=\sum_{k=1}^{n}\frac{k^2-\frac{1}{2}}{\left(k^2+k+\frac{1}{2}\right)\left(k^2-k+\frac{1}{2}\right)}=\sum_{k=1}^{n}\left(\frac{k-\frac{1}{2}}{\left(k^2-k+\frac{1}{2}\right)}-\frac{k+\frac{1}{2}}{\left(k^2+k+\frac{1}{2}\right)}\right)$

$=1-\frac{n+\frac{1}{2}}{n^2+n+\frac{1}{2}}=\frac{2n^2}{2n^2+2n+1}$

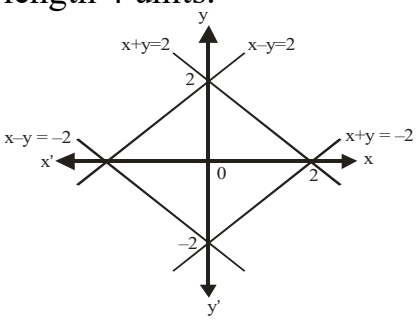
05. PQ and AB intersect inside the circle

Let $PR=a,RS=b,QS=c$ $\quad\therefore a(b+c)=3\times7\Rightarrow a=c$

06. $J=\int_0^{\infty}\frac{\sin(2025x)\cdot\cos(2024x)}{x}dx=\int_0^{\infty}\frac{\sin(4049x)+\sin x}{2x}dx$

07. $\left(2023+2024\frac{dy}{dx}\right)^{2026\times2029}=(2027)^{\lambda}\left(\frac{d^3y}{dx^3}\right)^3$

08. $2\ge\max\{|x-y|,|x+y|\}\Rightarrow|x-y|\le2$ and $|x+y|\le2$, which forms a square of diagonal length 4 units.



$\Rightarrow$ The area of the region is $\frac{1}{2}\times 4\times 4 = 8$ sq.units

This is equal to the area of the square of side length $2\sqrt{2}$.

09. Conceptual

10. Given $I=\int\frac{e^x}{e^{4x}+1}\,dx$ and $J=\int\frac{e^{-x}}{e^{-4x}+1}\,dx=\int\frac{e^{4x}.e^{-x}}{1+e^{4x}}\,dx=\int\frac{e^{3x}}{1+e^{4x}}\,dx$

$$\therefore I+J=\int\frac{e^x\left(1+e^{2x}\right)}{1+e^{4x}}\,dx$$

Put $e^x=t\Rightarrow e^x dx=dt$ $\quad\therefore I+J=\int\frac{1+t^2}{1+t^4}\,dx=\int\frac{1+\frac{1}{t^2}}{\left(t-\frac{1}{t}\right)^2+2}\,dt$

Put $t-\frac{1}{t}=y\Rightarrow dy=\left(1+\frac{1}{t^2}\right)dt$

$$\therefore I+J=\int\frac{dy}{y^2+2}=\frac{1}{\sqrt{2}}\tan^{-1}\frac{y}{\sqrt{2}}+C=\frac{1}{\sqrt{2}}\tan^{-1}\frac{t^2-1}{\sqrt{2}t}+C$$

$$I+J=\frac{1}{\sqrt{2}}\tan^{-1}\left(\frac{e^{2x}-1}{\sqrt{2}e^x}\right)+C$$

$$J-I=\int\frac{e^x\left(e^{2x}-1\right)}{1+e^{4x}}dx=\int\frac{e^x\left(1-e^{-2x}\right)}{e^{-2x}+e^{2x}}dx$$

Put $e^x=t\Rightarrow e^x dx=dt$

$$\therefore J-I=\int\frac{t^2-1}{1+t^4}\,dx=\int\frac{1-\frac{1}{t^2}}{\left(t+\frac{1}{t}\right)^2-2}\,dt$$

Put $t+\frac{1}{t}=y\Rightarrow dy=\left(1-\frac{1}{t^2}\right)dt$

$$\therefore J-I=\int\frac{dy}{y^2-2}=\frac{1}{2\sqrt{2}}ln\left|\frac{y-\sqrt{2}}{y+\sqrt{2}}\right|+C$$

$$J-I=\frac{1}{2\sqrt{2}}ln\left(\frac{e^{2x}-\sqrt{2}e^x+1}{e^{2x}+\sqrt{2}e^x+1}\right)+C$$

$$I+J=\frac{1}{\sqrt{2}}\tan^{-1}\left(\frac{e^{2x}-1}{\sqrt{2}e^x}\right)+C$$

So, we get $I=\frac{1}{2\sqrt{2}}\left[\tan^{-1}\left(\frac{e^{2x}-1}{\sqrt{2}e^{x}}\right)-\frac{1}{2}ln\left|\frac{e^{2x}-\sqrt{2}e^{x}+1}{e^{2x}+\sqrt{2}e^{x}+1}\right|\right]+C$

and $J=\frac{1}{2\sqrt{2}}\left[\tan^{-1}\left(\frac{e^{2x}-1}{\sqrt{2}e^{x}}\right)+\frac{1}{2}ln\left|\frac{e^{2x}-\sqrt{2}e^{x}+1}{e^{2x}+\sqrt{2}e^{x}+1}\right|\right]+C$

11. Given, $\sin^3 x=\frac{1}{2}\sin y$ and $\cos^3 x=\frac{1}{2}\cos y$

$\Rightarrow \sin^6 x+\cos^6 x=\frac{1}{4}\Rightarrow 1-\frac{3}{4}\sin^2 2x=\frac{1}{4}\Rightarrow \sin^2 2x=1\Rightarrow \cos 2x=0$

So, $x=\frac{\pi}{4},\frac{3\pi}{4},\frac{5\pi}{4},\frac{7\pi}{4}$ Now, we have $\tan y=\tan^3 x$

For each x, we get 4 values of y , Number of ordered pairs will be 16

12. Given, $\frac{x^2}{4}+\frac{y^2}{3}=1\Rightarrow$ Foci are $(\pm 1,0)$ and assume $P(\theta)$ as $\left(2\cos\theta,\sqrt{3}\sin\theta\right)$

Let Centroid be $(h,k)\Rightarrow h=\frac{2\cos\theta}{3},k=\frac{\sqrt{3}\sin\theta}{3}$

$\Rightarrow$ Locus of centroid of $\Delta PSS'$ will be $\frac{x^2}{4}+\frac{y^2}{3}=\frac{1}{9}$

13. $A^2=A\Rightarrow A=A^{-1}=I=A^T$; $B^2=I\Rightarrow B=B^{-1}$; $CC^T=I\Rightarrow C^T=C^{-1}$

(a) $(ABC)^{-1}=C^{-1}B^{-1}A^{-1}=C^TB=\left(AB^TC\right)^T$

(b) $\left|adj\left(2AB^{-1}C\right)\right|=\left|2AB^{-1}C\right|^2=64$

(c) $(ABC)^{-1}=(CBA)^{-1}\Rightarrow ABC=CBA\Rightarrow BC=CB$

(d) $adj\left(3A^{-1}BC^{-1}\right)=adj\left(3BC^{-1}\right)=9adj\left(BC^{-1}\right)$ $\left(C\left(B^T\right)^{-1}A\right)^T=B^{-1}C^T=BC^{-1}$

14. We use $\frac{x}{1-x^2}=\frac{1}{1-x}-\frac{1}{1-x^2}$ and $\frac{1}{x-1}-\frac{2}{x^2-1}=\frac{1}{x+1}$ to get $f(x)=\frac{x}{1-x}$

And $g(x)=\frac{1}{x-1}-\frac{2^{n+1}}{x^{2^{n+1}}-1}$

15. $E_1:\frac{x}{x-1}>0\Rightarrow x\in(-\infty,0)\cup(1,\infty)$, $E_2:-1\le ln\left(\frac{x}{x-1}\right)\le 1\Rightarrow x\in\left(-\infty,\frac{1}{1-e}\right)\cup\left[\frac{e}{e-1},\infty\right)$

As range of $\frac{x}{x-1}$ is $R^+-\{1\}$

So, Range of f is $R-\{0\}$ and Range of g is $\left[\frac{-\pi}{2},\frac{\pi}{2}\right]-\{0\}$

16. Key: b

Sol:(P) $P(A|C)+P(B|C)-P((A\cap B)|C)=\dfrac{P(A\cap C)+P(B\cap C)-P(A\cap B\cap C)}{P(C)}$

$=\dfrac{P((A\cup B)\cap C)}{P(C)}=P(A\cup B|C)$

(Q) $P((A\cap \overline{B})|C)+P((A\cap B)|C)=\dfrac{P(A\cap \overline{B}\cap C)+P(A\cap B\cap C)}{P(C)}$

$=\dfrac{P(A\cap C)}{P(C)}=P(A|C)$

(R) $\dfrac{1-P(A)-P(B\cup C)+P(A\cap(B\cup C))}{1-P(B\cup C)}=\dfrac{1-P(A\cup(B\cup C))}{1-P(B\cup C)}=\dfrac{P(\overline{A\cup(B\cup C)})}{P(\overline{B\cup C})}$

$=\dfrac{P(\overline{A}\cap(\overline{B}\cap\overline{C}))}{P(\overline{B}\cap\overline{C})}=P(\overline{A}|(\overline{B}\cap\overline{C}))$

(S) $P(\overline{A}\cup\overline{B})+P(A\cap B)=P(\overline{A\cap B})+P(A\cap B)=1$

17. (I) Required probability=$\dfrac{1}{2}$

(II) Required probability=$\left(\dfrac{1}{2}\right)^3+\left(\dfrac{1}{2}\right)^4+\left(\dfrac{1}{2}\right)^4+\left(\dfrac{1}{2}\right)^4+\left(\dfrac{1}{2}\right)^4-\left(\dfrac{1}{2}\right)^7=\dfrac{47}{2^7}$

(III) Required probability=$\dfrac{4}{2^7}$

(IV) There are five cases 0,1,2,3 or 4 wins, Required probability=$\dfrac{1+7+15+10+1}{2^7}=\dfrac{34}{2^7}$

18. Line is $\dfrac{x-1}{1}=\dfrac{y-1}{-1}=\dfrac{z+1}{1}$;

P_1 will be $1(x-1)-1(y-1)=0\Rightarrow x-y=0$

P_2 can be expressed as $(x+y-2)+\lambda(y+z)=0$ which should pass through $\overrightarrow{v_2}=-\hat{j}$.

$\Rightarrow(-3)+\lambda(-1)=0\Rightarrow\lambda=-3$

So, P_2 will be $x-2y-3z-2=0$ $\quad a=\dfrac{1}{\sqrt{2}}$ Let B be $(1+t,1-t,t-1)$

$\overrightarrow{AB}$ is parallel to the plane $2x-y-3z=1\Rightarrow 2(2+t)+1(-1-t)+3(t)=0$

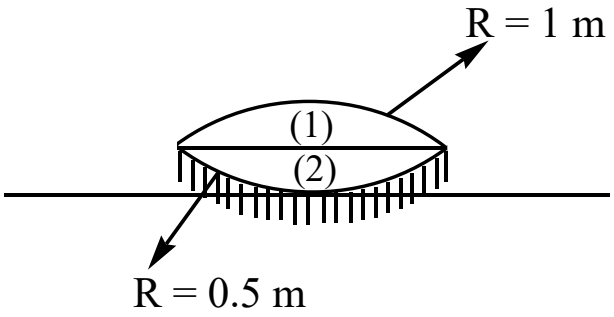
$\Rightarrow 4t=-3\Rightarrow t=\dfrac{-3}{4}$ $\quad$ So, $b=\dfrac{1}{4}$

Angle between $x-y=0$ and $x-2y-3z-2=0$ is

$\cos^{-1}\left(\dfrac{3}{2\sqrt{7}}\right)=\sin^{-1}\left(\dfrac{\sqrt{19}}{2\sqrt{7}}\right)=\tan^{-1}\left(\dfrac{\sqrt{19}}{3}\right)$ $p=19,q=3$

PHYSICS

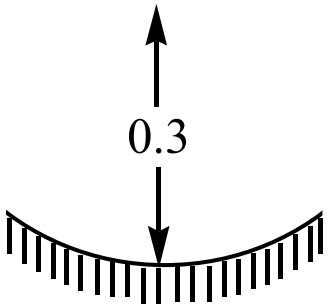
19. $\frac{1}{f_1}=(1.5-1)\left(\frac{1}{1}\right)=0.5m^{-1}$ $\frac{1}{f_2}=\left(\frac{4}{3}\right)\left(\frac{1}{0.5}\right)$



$=\frac{1}{3}\times\frac{1}{0.5}=\frac{2}{3}m^{-1}$

Power of the system

$P=2P_1+2P_2+P_m=2\times0.5+2\times\frac{2}{3}+\frac{1}{0.25}=1+\frac{4}{3}+4=\frac{3+4+12}{3}=\frac{19}{3}$



Focal length of equivalent mirror $=-\frac{3}{19}m$

$\frac{1}{v}+\frac{1}{-0.3}=\frac{1}{-3/19}$ $\frac{1}{v}=\frac{1}{0.3}-\frac{19}{3}=\frac{10}{3}-\frac{19}{3}=\frac{-9}{3}=-3$

$v=-\frac{1}{3}m \Rightarrow d=-\frac{3}{10}+\frac{1}{3}=\frac{1}{30} \Rightarrow \frac{1}{30}=\frac{x}{90}\Rightarrow x=3m$

20. Initially let number of molecules of $A=N_A$ and that of $B=N_B$

$\frac{\lambda N_A}{\lambda N_A+2\lambda N_B}=\frac{25}{100}$ $\therefore 100\lambda N_A=25\lambda N_A+50\lambda N_B\Rightarrow N_A=\frac{2}{3}N_B$

Now, $N_A=N_Ae^{-\lambda t}=\frac{2}{3}N_Be^{-\lambda t}$

$N_B=N_Be^{-2\lambda t}=\frac{2}{3}N_Be^{-\lambda t}$

$\frac{\frac{2}{3}\lambda N_Be^{-\lambda t}}{\frac{2}{3}\lambda N_Be^{-\lambda t}+2\lambda N_Be^{-\lambda t}}=\frac{75}{100}=\frac{3}{4}$

$$\Rightarrow \frac{1}{1+3e^{-\lambda t}} = \frac{3}{4} \quad \Rightarrow 1 = 9e^{-\lambda t} \quad \Rightarrow e^{\lambda t} = 9$$

So, $t = \frac{\ln(9)}{\lambda} hr = \frac{2\ln(3)}{\ln(3)} = 2hr$

21. The torque due to the viscous liquid on the top of the solid cylinder

$$\tau_1 = \int_0^R \eta 2\pi r dr \frac{\omega r}{h} r \qquad \tau_1 = 2\pi\eta \frac{\omega}{h} \int_0^R r^3 dr$$

$$\tau_1 = 2\pi\eta \frac{\omega}{h} \frac{R^4}{4} = \frac{\pi\eta R^4}{2h} \qquad \text{.....(i)}$$

The torque due to the viscous liquid on the curved surface of the solid cylinder

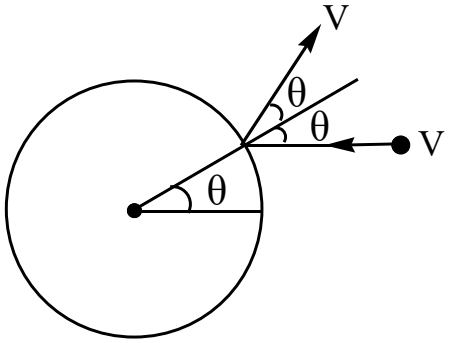
$$\tau_2 = \eta 2\pi Rl \frac{\omega R}{h} R \quad \tau_2 = 2\pi\eta R \times 5R \frac{\omega R^2}{h}$$

$$\tau_2 = 10\pi\eta \frac{\omega R^4}{h} \qquad \text{.....(ii)}$$

Hence the net torque required, $\tau = \tau_1 + \tau_2$

$$\tau = \left(\frac{\pi\eta\omega R^4}{2h}\right) + \left(\frac{10\pi\eta\omega R^4}{h}\right) \quad \tau = \frac{21\pi\eta R^4}{2h}$$ Hence, K = 21

22. In the frame of the heavy cylinder the particle comes in with speed V and then bounces off.



$$f = \int_{-\frac{\pi}{2}}^{+\frac{\pi}{2}} 2mv\cos^2\theta (Rd\theta\cos\theta) lvn \qquad \frac{f}{l} = \frac{8}{3}$$

23. The frequency of sound received by the reflector car B,

$$f_1 = \left(\frac{v+\omega+u}{v+\omega-u}\right) f \qquad \text{.....(i)}$$

The frequency of the reflected sound wave received by the driver of car A

$$f_2 = \left(\frac{v-\omega+u}{v-\omega-u}\right) f_1$$

$$f_2 = \left(\frac{v-\omega+u}{v-\omega-u}\right)\left(\frac{v+\omega+u}{v+\omega-u}\right) f \qquad \text{.....(ii)}$$

$$f_2 = \left(\frac{330-5+25}{330-5-25}\right)\left(\frac{330+5+25}{330+5-25}\right)310 \qquad \text{.....(ii)}$$

$$f_2 = \frac{350}{300}\times\frac{360}{310}\times 310 \qquad f_2 = 420\ Hz$$

24. Time period of Earth's revolution around the Sun

$$T = \frac{2\pi}{\sqrt{GM_S}}R^{3/2} \qquad \text{.....(i)}$$

Time period of the double stars binary system

$$T_b = \frac{2\pi\times(36R)^{3/2}}{\sqrt{G(4M_S+12M_S)}} \quad T_b = \frac{2\pi\times(6)^3R^{3/2}}{\sqrt{G16M_S}}$$

$$T_b = \frac{216}{4}\times\frac{2\pi R^{3/2}}{\sqrt{GM_S}} \qquad T_b = 54T \qquad n = 54$$

25. ${}_1^2H + {}_1^2H \longrightarrow {}_1^3H + {}_1^1P$

The Q-value of the reaction

$Q = (2.8\times 3) - (1\times 4) \qquad \Rightarrow Q = 4.4\ MeV$

Using conservation of momentum

$\vec{P}_1 + \vec{P}_2 = 0 \Rightarrow \vec{P}_1 = -\vec{P}_2 \qquad P_1^2 = P_2^2 \qquad 2m_1K_1 = 2m_2K_2$

$K_1 = 3K_2 \qquad \text{....(i)}$

Also, $K_1 + K_2 = Q$

$K_1 + K_2 = 4.4 \qquad \text{....(ii)}$

$4K_2 = 4.4 \Rightarrow K_2 = 1.1MeV \ and \ K_1 = 3.3MeV$

Hence $(10K_1 + 20K_2) = (10\times 3.3) + (20\times 1.1) = (33+22) = 55MeV$

26. $C_{eq} = \frac{C_1C_2}{C_1+C_2} = \frac{10\times 40}{10+40} = 8\mu F \qquad \text{....(i)}$

Now, $\frac{1}{C_{eq}} = \frac{1}{C_1} + \frac{1}{C_2} \qquad \frac{dC_{eq}}{C_{eq}^2} = \frac{dC_1}{C_1^2} + \frac{dC_2}{C_2^2}$

$$\frac{dC_{eq}}{C_{eq}} = \left(\frac{dC_1}{C_1^2} + \frac{dC_2}{C_2^2}\right)C_{eq}$$

$$\frac{dC_{eq}}{C_{eq}} = \left(\frac{0.10}{100} + \frac{0.20}{1600}\right)\times 8 = \frac{1.80}{200} = 9\times 10^{-3} \qquad \text{.....(ii)}$$

Now, $U = \frac{1}{2}C_{eq}V^2 \quad \frac{dU}{U} = \frac{dC_{eq}}{C_{eq}} + \frac{2dV}{V} = 9\times 10^{-3} + 2\times\frac{0.02}{8} = 14\times 10^{-3}$

Hence, $\frac{dU}{U\times 100} = 14\times 10^{-3}\times 100 = 1.4\% \qquad \Rightarrow n = 1.4$

Hence, $5n = 7$

27. $\frac{dN_B}{dt} = \lambda_1 N_A - \lambda_2 N_B$

at $t = t_0, \frac{dN_B}{dt} = 0, N_B = \frac{3N_0}{2} \Rightarrow 2N_0\lambda_1 e^{-\lambda_1 t_0} - \lambda_0 \frac{3N_0}{2} = 0$

$t_0 = \frac{1}{\lambda_1} \ln \frac{4}{3} \frac{\lambda_1}{\lambda_2}$

28. Using work-energy theorem till string become taut

Work done by qE = Change in $K.E$.

$\Rightarrow qEI = \frac{1}{2}mv^2 - 0 \qquad \Rightarrow v = \sqrt{\frac{2qEl}{2m}}$

When string becomes taut an impulse is received and speed of the bod becomes

$v\cos 30^0 = \sqrt{\frac{3qEl}{2m}}$

Further using work energy theorem speed of the bob becomes

$v' = \sqrt{\frac{5qEl}{2m}} \quad \therefore \text{time period} = \frac{2\pi l}{v'} = 2\pi\sqrt{\frac{2ml}{5qE}}$

After 1st half revolution in magnetic field work done by qE = Change in $K.E$.,

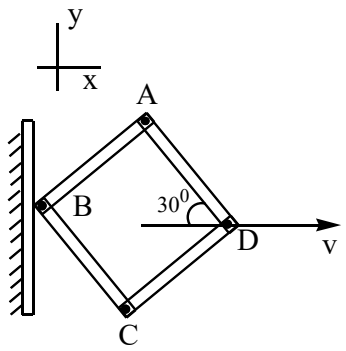
from lowermost to uppermost point. $qE.2l = \frac{mv_1^2}{2} - \frac{m}{2}\frac{5qEl}{2m} \qquad v_1 = \sqrt{\frac{13qEl}{2m}}$

$\therefore$ Time period of 2nd revolution $= \frac{2\pi l}{v_1} = \pi\sqrt{\frac{2ml}{13qE}}$

$\therefore$ Angular speed of 1st revolution $= \sqrt{\frac{5qE}{2ml}}$

$\therefore$ Angular speed of 2nd revolution $= \sqrt{\frac{13qE}{2ml}} \quad \therefore$ ABC

29. $v_x = v/2 \qquad v_y = v_x \cot 30^0$



$v_y = \frac{v}{2}\sqrt{3} \qquad v_A = v$

30. Direction of magnetic field at P due to $CD = \hat{i} \times \frac{\left(\hat{j} + \hat{k}\right)}{\sqrt{2}} \Rightarrow \frac{k - \hat{j}}{\sqrt{2}}$

Direction of magnetic field at P due to $EF = -\hat{i} \times \dfrac{\left(\hat{j}+\hat{k}\right)}{\sqrt{2}} \Rightarrow \dfrac{-k-\hat{j}}{\sqrt{2}}$

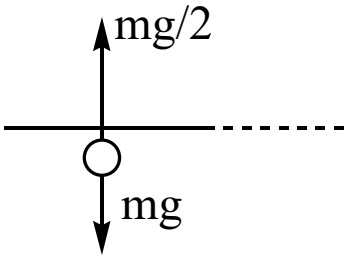
Hence direction will be along $-\hat{j}$

Direction of magnetic field at 'O' due to upper frame along $-\hat{i}$

Direction of magnetic field at O due to horizontal frame will be along $-\hat{j}$

31. Travelling harmonic wave is sin or cos function of $(ax \pm bt)$.

32. $ma = mg - \dfrac{mg}{2} - 6\pi\eta r v$



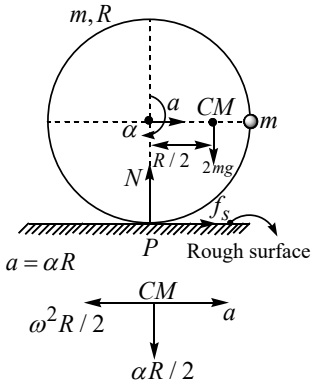
$ma = 6\pi\eta r v_T - 6\pi\eta r v \quad \dfrac{dv}{dt} = \dfrac{6\pi\eta r}{m}(v_T - v) \quad \dfrac{dv}{v_T - v} = \dfrac{6\pi\eta r}{m}dt \quad \dfrac{v_T - v}{v_T} = e^{-\frac{6\pi\eta r}{m}t}$

$v = v_T\left(1 - e^{\frac{6\pi\eta r}{m}t}\right) \qquad v = v_T\left(1 - e^{-\frac{6\pi\eta r}{m}t}\right) \quad \dfrac{1}{2} = 1 - e^{-\frac{6\pi\eta r}{m}t} \quad -\dfrac{6\pi\eta r t}{m} = ln2 \; t = \dfrac{mln2}{6\pi\eta r}$

33. $I_P = 2mR^2 + 2mR^2 = 4mR^2$, Using conservation of energy

$mg\dfrac{R}{2} = \dfrac{1}{2} \times 4mR^2\omega^2 \Rightarrow \omega = \sqrt{\dfrac{g}{4R}}$ .....(i)

Now, $\tau_P = I_P\alpha \Rightarrow 2m(g + \omega^2 R)\dfrac{R}{2} = 4mR^2\alpha$



$\Rightarrow \left(g + \dfrac{g}{4}\right) = 4R\alpha \Rightarrow \alpha = \dfrac{5g}{16R}$ ....(ii)

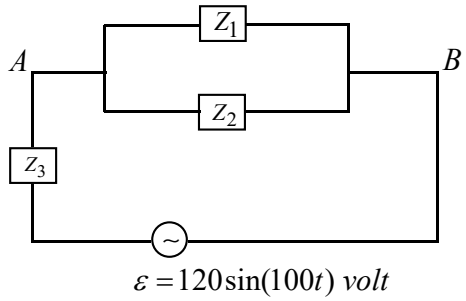
$2mg - N = 2m\dfrac{\alpha R}{2} \Rightarrow 2mg - N = \dfrac{5mg}{16} \qquad N = \dfrac{27mg}{16} = 13.5\ N$

$f_B = 2m\left(a - \dfrac{\omega^2 R}{2}\right) = 2m\left(\dfrac{5g}{16} - \dfrac{g}{8}\right) = \dfrac{3mg}{8} = 3N$

$$F_h = m(\alpha R - \omega^2 R) = m\left(\frac{5g}{16} - \frac{g}{4}\right) = \frac{mg}{16} = 0.5\ N$$

$$mg - F_v = m\alpha R \Rightarrow F_v = mg - \frac{5mg}{16} = \frac{11mg}{16} = 5.5\,N$$

34.



$\varepsilon = 120\sin(100t)\ volt$

$$Z_1 = R_1 + \omega L i = 20 + 20i$$

$$Z_2 = R_2 - \frac{1}{\omega C} i = 20 - \frac{1}{100 \times 5 \times 10^{-4}} i = 20 - 20i \qquad \frac{1}{Z'} = \frac{1}{Z_1} + \frac{1}{Z_2}$$

$$\frac{1}{Z'} = \frac{1}{20(1+i)} + \frac{1}{20(1-i)} \quad \frac{1}{Z'} = \frac{(1-i)}{40} + \frac{(1+i)}{40} \Rightarrow \frac{1}{Z'} = \frac{2}{40} \qquad \Rightarrow Z' = 20\Omega$$

Hence, $Z = Z' + Z^3 = 20 + (10 + 30i) \quad \Rightarrow Z = 30(1+i)$

The peak current through the source, $I_0 = \frac{\varepsilon_0}{|Z|} = \frac{120}{30\sqrt{2}} = 2\sqrt{2}\ Amp$

The instantaneous current through the source, $i = 2\sqrt{2}\sin\left(100t - \frac{\pi}{4}\right)$

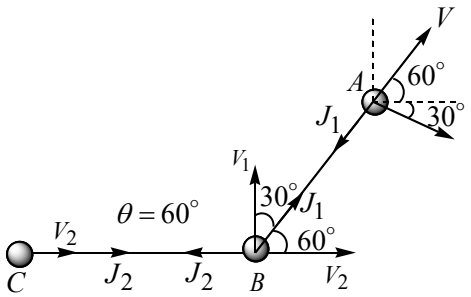
The instantaneous potential drop between A and B, $v' = 40\sqrt{2}\sin\left(100t - \frac{\pi}{4}\right)$

The peak current through the resistor R_2, $I_2 = \frac{40\sqrt{2}}{20\sqrt{2}} = 2\ amp$

The average power dissipated in the resistor R_3, $P_3 = I_{rms}^2 R_3$

$$\Rightarrow \qquad P_3 = \left(\frac{I_0}{\sqrt{2}}\right)^2 R_3 = (2)^2 \times 10 = 40\ watt$$

35.



$$v_1 \cos 30^0 + v_2 \cos 60^0 = v$$

$$v_1 \frac{\sqrt{3}}{2} + \frac{v_2}{2} = v$$

$\sqrt{3}v_1 + v_2 = 2v$ .....(i)

Using conservation of momentum along x and y axes

$mu = m(v\cos 60^0 + u\cos^2 30^0) + 2mv_2$

$u = \frac{v}{2} + \frac{3u}{4} + 2v_2 \Rightarrow 8v_2 + 2v = u$ .....(ii)

Also, $mv_1 + m(v\cos 30^0 - u\cos 30^0 \cos 60^0) = 0$

$v_1 + \frac{v\sqrt{3}}{2} = \frac{\sqrt{3}}{4}u \Rightarrow 4v_1 + 2\sqrt{3}v = \sqrt{3}u$ ……(iii)

Solving equations (ii) and (iii), we get

$4v_1 + \sqrt{3}(u - 8v_2) = \sqrt{3}u \quad 4v_1 = 8\sqrt{3}v_2 \Rightarrow v_1 = 2\sqrt{3}v_2$

Solving equations (i) and (ii), we get

$\sqrt{3}v_1 + v_2 = u - 8v_2 \quad \Rightarrow 6v_2 + 9v_2 = u$

$\Rightarrow v_2 = \frac{u}{15} = \frac{30}{15} = 2\ m/s$ and $v_1 = 2\sqrt{3}v_2 = 4\sqrt{3}\ m/s$

Hence, $v_B = \sqrt{v_1^2 + v_2^2} = \sqrt{48+4} = 2\sqrt{13} = 7.2\, m/s \ \ v_C = v_2 = 2\ m/s$

Now, $J_1\cos 60^0 = 2mv_2\ A\ J_1 = 4mv_2 = 16\ N-s$

$J_2 = mv_2 = 2\times 2 = 4\ N-s$

36. $E_1 = \frac{\lambda_1}{2\pi\varepsilon_0 r_1}$ and $E_2 = \frac{\lambda_2}{2\pi\varepsilon_0 r_2}$

$F_1 = P\left|\frac{dE_1}{dr_1}\right| = P\left|\frac{\lambda_1}{2\pi\varepsilon_0 r_1^2}\right|$ and $F_2 = P\left|\frac{dE_2}{dr_2}\right| = P\left|\frac{\lambda_2}{2\pi\varepsilon_0 r_2^2}\right|$

(P) $F_1 = \frac{P2\lambda\times 4}{2\pi\varepsilon_0 d^2} = \frac{4\lambda P}{\pi\varepsilon_0 d^2}$ and $F_2 = \frac{P\lambda 4}{2\pi\varepsilon_0 d^2} = \frac{2\lambda P}{\pi\varepsilon_0 d^2}$

Hence, $F = F_1 - F_2 = \frac{2\lambda P}{\pi\varepsilon_0 d^2}$

(Q) $F_1 = \frac{4\lambda P}{\pi\varepsilon_0 d^2}$ and $F_2 = \frac{2\lambda P}{\pi\varepsilon_0 d^2}$

Hence, $F = F_1 - F_2 = \frac{2\lambda P}{\pi\varepsilon_0 d^2}$

(R) $F_1 = \frac{P\lambda\times 16}{2\pi\varepsilon_0 d^2} = \frac{8\lambda P}{\pi\varepsilon_0 d^2}$ and $F_2 = \frac{P2\lambda\times 9}{2\pi\varepsilon_0 d^2} = \frac{9\lambda P}{\pi\varepsilon_0 d^2}$

Hence, $F = F_1 + F_2 = \frac{17\lambda P}{\pi\varepsilon_0 d^2}$

(S) $F_1 = \frac{8P\lambda}{\pi\varepsilon_0 d^2}$ and $F_2 = \frac{9P\lambda}{\pi\varepsilon_0 d^2}$ Hence, $F = F_2 - F_1 = \frac{\lambda P}{\pi\varepsilon_0 d^2}$

CHEMISTRY

37. Energy of photon $= \frac{1240}{6.2} = 200eV$

KE_{max} of electron = 200 – 50 = 150 eV

$\lambda = \sqrt{\frac{150}{150}} = 1\text{Å} = 10^{-10}m; \qquad P = \frac{h}{\lambda}$

$\frac{dP}{d\lambda} = -\frac{h}{\lambda^2}; \quad d\lambda = \frac{-\lambda^2}{h}dP$

$d\lambda = \frac{10^{-10} \times 10^{-10} \times 1.8 \times 10^{15}}{6 \times 10^{-34}} = 30 \text{ m}$

38. $C(graphite) + 2H_2(g) \rightarrow CH_4(g)$

$\Delta S_{rxn} = [186.2] - [6 + 130.6 \times 2] \qquad = -81 \text{ Joule}$

$\Delta G_{Rxn} = \Delta H_{Rxn} - T\Delta S_{rxn} = -75 \times 10^3 + 81 \times 300 = -50.7 \times 10^3$

$\Delta G_{Rxn} = -T\Delta S_{univ}$

$\Delta S_{univ} = \frac{50.7 \times 10^3}{300} \qquad = 169$

39. $Mg(OH)_2 \rightleftharpoons \underset{x}{Mg^{+2}} + \underset{2x+3y}{2OH^-}$ $\quad K_{SP}$ of $Mg(OH)_2 > K_{SP}$ of $Al(OH)_3$

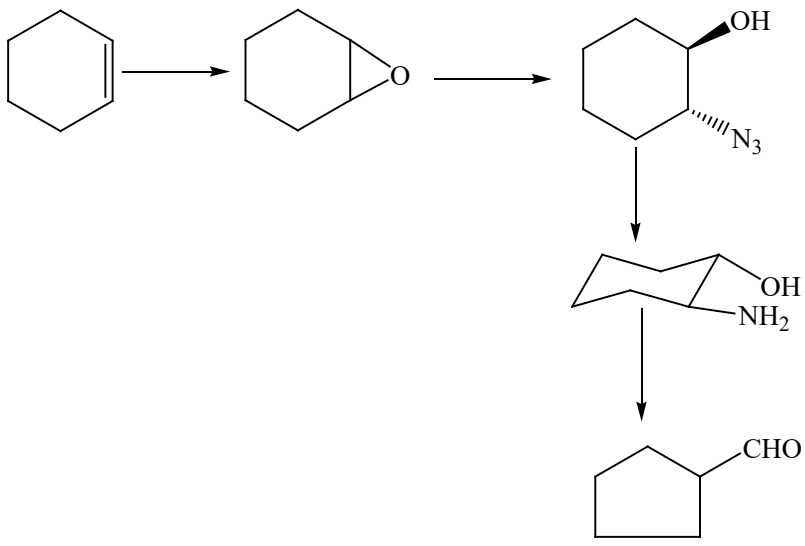
$Al(OH)_3 \rightleftharpoons \underset{y}{Al^{3+}} + \underset{3y+2x}{3OH^-}$ $\qquad$ so, $\underset{2x+3y=2x}{x >> y}$

$\because 4 \times 10^{-12} = [Mg^{2+}][OH^-]^2 \qquad = x \times (2x)^2$

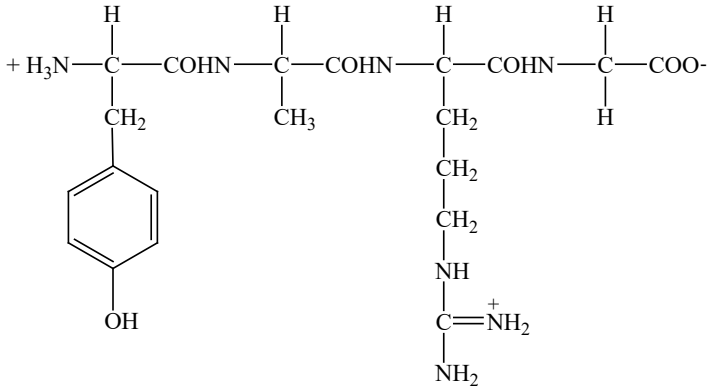
$1 \times 10^{-33} = [Al^{3+}][OH^-]^3 \qquad 1 \times 10^{-33} = (y)(2x)^3$

$1 \times 10^{-33} = y \times (10^{-4} \times 2)^3 \; y = \frac{10^{-21}}{8}$ so $\frac{x}{y} = 8 \times 10^{17} \; 8 \times 10^{17} \times 10^{-16} = 80$

40.

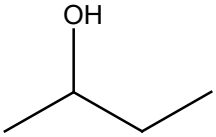
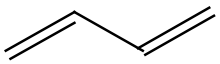


41.



Net charge = (-) + (+) + (0) + (+) = +1

42. $A = CH_3 - C \equiv C - CH_3$

B=  C=  X = 2, y = 54

43. $4FeCl_3 + 3K_4\left[Fe(CN)_6\right] \rightarrow Fe_4\left[Fe(CN)_6\right]_3 + 12KCl$

44. Hint: Single N – N bond is weaker than the single P –P bond.

45. (A) At $400k\ K_P = 50$ At $500k\ K_P = 1700$

As temperature increases K_P value also increases so the reaction is endothermic.

(C) $Q_P = \dfrac{P_{NO_2}^2}{P_{N_2O_4}} = \dfrac{20 \times 20}{2} = 200$ At 400K $K_p = 50$ $Q_P > K_P$

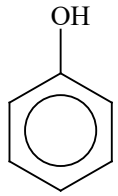
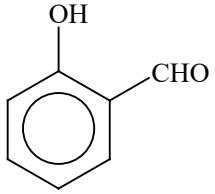
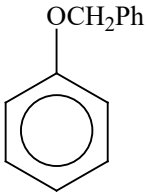
So the reaction goes to backward direction. It means more $N_2O_4(g)$ will be obtained

(D) Entropy increases because $\Delta n_g = +ve$

46. $P_{AB} = \dfrac{75+22}{2} = 48.5\ torr$ $P_{AC} = \dfrac{75+10}{2} = 42.5\ torr$

$P_{BC} = \dfrac{22+10}{2} = 16\ torr$ $P_{ABC} = \dfrac{75+22+10}{3} = 35.67\ torr$

47.

P =  Q =  S = 

48. Conceptual

49. i) $Na_2S_2O_5 : S - S\ linkage$, ii) $C_3O_2 : O = C = C = C = O$

50. (i) $PbS + 4H_2O_2 \rightarrow PbSO_4 + 4H_2O(x = 4)$

(ii) $2MnO_4^- + 6H^+ + 5H_2O_2 \rightarrow 2Mn^{2+} + 8H_2O + 5O_2(y = 5)$

(iii) $AlCl_3 + 3H_2O \rightarrow Al(OH)_3 + 3HCl(p = 1)$

(iv) $Ca_3N_2 + 6H_2O \rightarrow 3Ca(OH)_2 + 2NH_3(q = 2)$

51. Volume of $H_2SO_4 = 10ml$, Wt of H_2SO_4 solution = 14.7 gram

For reaction (ii)

Molarity of NaOH $= \frac{8\times10}{40} = 2$

$[Mole\times n.f]_{H_2SO_4} = [Mole\times n.f]_{NaOH}$

Mole $\times 2 = 2\times20\times10^{-3}\times1$ Mole $= 20\times10^{-3}$

mole of H_2SO_4 reacted in reaction-II $= 20\times10^{-3}$

From Reaction (iii)

Mole of $I_2 = \frac{1}{2}\times$ mole of NaI $= \frac{1}{2}\times\frac{03}{150} = \frac{0.6}{150}$

From Reaction (iii)

Mole of I_2 = mole of $Cl_2 = \frac{0.497}{71} = 7\times10^{-3}$

Total mole of I_2 evolved $= 8\times10^{-3}$, Mole of $KI = 2\times mole\ of\ I_2 = 16\times10^{-3}$

Wt of KI = $16\times10^{-3}\times166 = 2656\times10^{-3}\ gram$

% purity of $KI = \frac{2.656}{10}\times100 = 26.56\%$

Mole of H_2SO_4 reacted in reaction (i) $= 10\times10^{-3}$

Total mole of $H_2SO_4 = 30\times10^{-3}$

Molarity of $H_2SO_4 = \frac{30\times10^{-3}}{10}\times1000$ $M = \frac{\%\left(\frac{Wt}{Wt}\right)\times10\times d}{98}$

$3\times98 = \%\left(\frac{Wt}{Wt}\right)\times10\times1.47$ $\%\left(\frac{Wt}{Wt}\right) = 20$

From reaction (iii) $[millimole\times n.f]_{I_2} = [M\times n.f\times V(ms)]_{Na_2S_2O_3}$

$1\times2 = M\times1\times20$ M = 0.1

52. Conceptual

53. (i) ClF_3: 2 lone pairs near central atom, trigonal bipyramidal geometry

(ii) I_3^-: 3 lone pairs near central atom, trigonal bipyramidal geometry

(iii) I_3^+: 2 lone pairs near central atom, tetrahedral

(iv) XeF_4: 2 lone pairs near central atom, octahedral

54. (i) $\left[Cr(NH_3)_6\right]^{3+}: Cr^{3+}: d^3: para: 3\ unpaired\ electrons$

(ii) $\left[Ni(NH_3)_6\right]^{2+}: Ni^{2+}: 3d^8: para: 2 unpaired\ electrons$

(iii) $\left[Ru(H_2O)_6\right]^{3+}: Ru^{3+}: 4d^5: low\ spin: para: 1\ unpaired\ electron$

(iv) $\left[CrF_6\right]^{3-}: Cr^{3+}: d^3: para: 3\ unpaired\ electrons$